GLOBE SCOTT MOTORS · FREE ENGINEERING RESOURCE

Custom Motor Specification Freeze Checklist

For OEM design, purchasing and quality teams preparing a custom PMDC, geared or BLDC motor project

Complete every item before samples are built and record the agreed value and document revision beside it. Any item left open is a likely cause of a re-sample. Items in Steps 1 and 2 should be frozen before design starts.

Step 1 — Load and duty cycle (freeze before design)

- ☐ Load torque at the output shaft in Nm for every segment of the machine cycle, including breakaway and acceleration.
- ☐ Output speed in rpm at the working load, with the permitted tolerance (for example ±5 %).
- ☐ Full cycle profile: on-time, off-time and starts per hour; state the IEC 60034-1 duty type (S1, S2 or S3) where known.
- ☐ RMS torque over the cycle calculated and agreed: Trms = square root of [(sum of $T^2 \times t$) / (sum of t)].
- ☐ Peak torque and its duration, including shock loads, jams and stall events.
- ☐ Ambient temperature at the motor itself, not the room, including summer peaks of 45–50 °C where relevant.

Step 2 — Electrical supply and control (freeze before design)

- ☐ Supply voltage at the motor terminals under load as minimum / nominal / maximum (example: 21 V / 24 V / 28.8 V on a 24 V lead-acid pack).
- ☐ Supply type: battery, rectified mains or electronic controller; note ripple and PWM frequency where a drive is used.
- ☐ Controller current limit and whether the motor is reversed or braked at full voltage.
- ☐ Cable length and size between controller and motor, so that voltage drop is included in the minimum voltage.
- ☐ Thermal protection required: none, bimetal switch, PTC thermistor or temperature sensor.

Step 3 — Mechanical interface (freeze before samples)

- ☐ Mounting type: IEC B3 foot, B5 flange or B14 face, with flange PCD, spigot diameter and hole details.
- ☐ Shaft diameter, extension length, tolerance, keyway or flat, and any second shaft end.
- ☐ Maximum envelope: length, diameter and clearances to neighbouring parts, supplied as a 2D drawing or STEP model.
- ☐ Gear ratio, output shaft orientation and permitted backlash, if geared.
- ☐ Overhung and axial load on the output shaft from pulleys, sprockets or pinions.
- ☐ Mounting orientation: horizontal, shaft-up or shaft-down.

Step 4 — Connections, options and environment (freeze before samples)

- ☐ Lead exit position, lead length, wire size and sleeving.
- ☐ Connector part number and mating half, and who supplies each.
- ☐ Options: electromagnetic brake (holding torque, coil voltage), encoder (resolution, output type), EMI suppression.
- ☐ IP rating and exposure: water, dust, washdown, monsoon humidity, chemicals.
- ☐ Insulation class and any special materials, coatings or marking.

Step 5 — Acceptance tests and approvals (freeze before samples)

- [ ] List of tests: no-load speed and current, load-point speed and current, temperature rise, insulation resistance, hi-pot, noise if required.
- [ ] Conditions for each test: supply voltage, load point, winding temperature (cold or hot) and instrument.
- [ ] Pass/fail tolerances written down and signed by both parties.
- [ ] Sample quantity covering bench, endurance, in-machine and any destructive tests.
- [ ] Approvers named, design review dates booked, and third-party or RDSO inspection planned where applicable.

Step 6 — Change control after the freeze

- [ ] One revision-controlled specification held by a single technical owner on the buyer's side.
- [ ] Every change raised as a written change request with reason and requested date.
- [ ] Impact note agreed before acceptance: cost, re-test scope and schedule effect.
- [ ] Long-lead parts released as soon as their governing parameters are frozen: magnets, gear sets, brakes, encoders, connectors.

When Each Item Must Be Frozen

Specification item	Freeze before	Typical effect of a late change
Load torque, speed, duty cycle	Design	Frame or stack change → new samples
Voltage range at terminals	Design	Re-wind → new samples and tests
Gear ratio	Design	New gear sets → bought-out lead time
Mounting, flange, envelope	Samples	End shield, adaptor or new tooling
Shaft details	Samples	Machining change, repeat fit checks
Brake, encoder, options	Samples	Length, wiring and thermal re-check
Leads and connectors	Samples	Sealing and ingress re-test
Acceptance tests	Samples	Validation cannot close
Approvals and inspections	Validation	Inspection slot re-booked

Values shown are examples. Agree actual figures, tolerances and test conditions with your motor manufacturer against your machine's real requirements, and keep this checklist with the signed specification.

Need help sizing or replacing a motor?
Globe Scott Motors has manufactured electric motors in Gujarat since 1985 and today builds PMDC, BLDC and geared motors, PMDC transaxles and railway signalling motors. Share your load, duty cycle and envelope at www.globescott.net/contact-us/

General engineering guidance only. Always follow the equipment manufacturer's instructions, applicable standards and site safety rules (isolation and lock-out/tag-out) before working on electrical or rotating equipment.